

AI-Driven System for Oil Spill Identification and Monitoring

Project Statement:

Oil spills pose a serious threat to marine ecosystems, coastal regions, and local economies. Traditional detection methods such as satellite image manual inspection or physical patrolling are time-consuming, labor-intensive, and often delayed. This project aims to develop an AI-powered oil spill detection system using machine learning and satellite imagery to identify and localize oil spills efficiently and accurately. Leveraging deep learning models such as CNNs or U-Net, the system will analyze satellite images to detect oil spill patterns and generate segmentation maps for real-time monitoring. This approach helps in early intervention, reducing environmental damage and supporting emergency response.

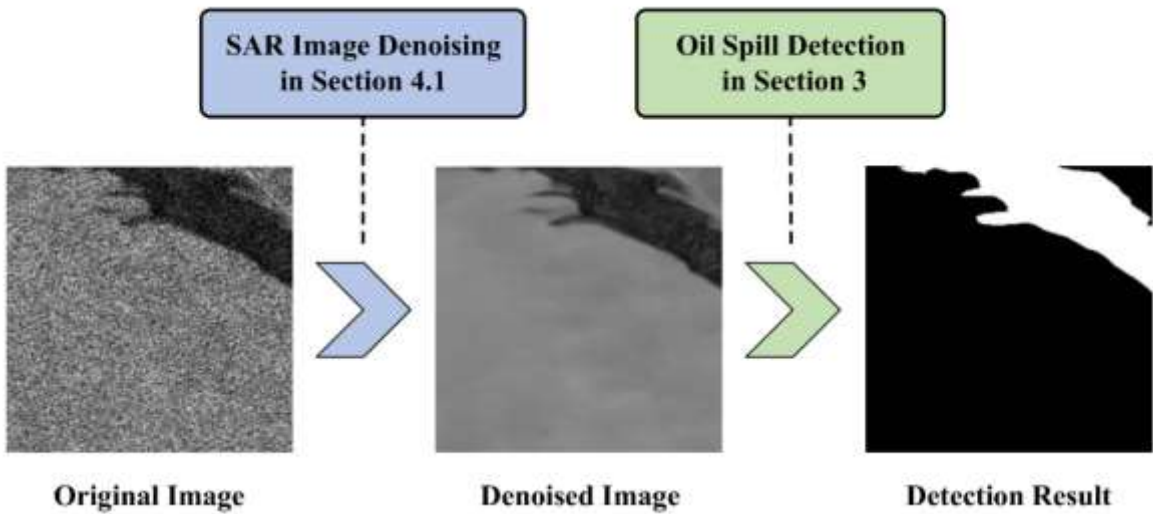
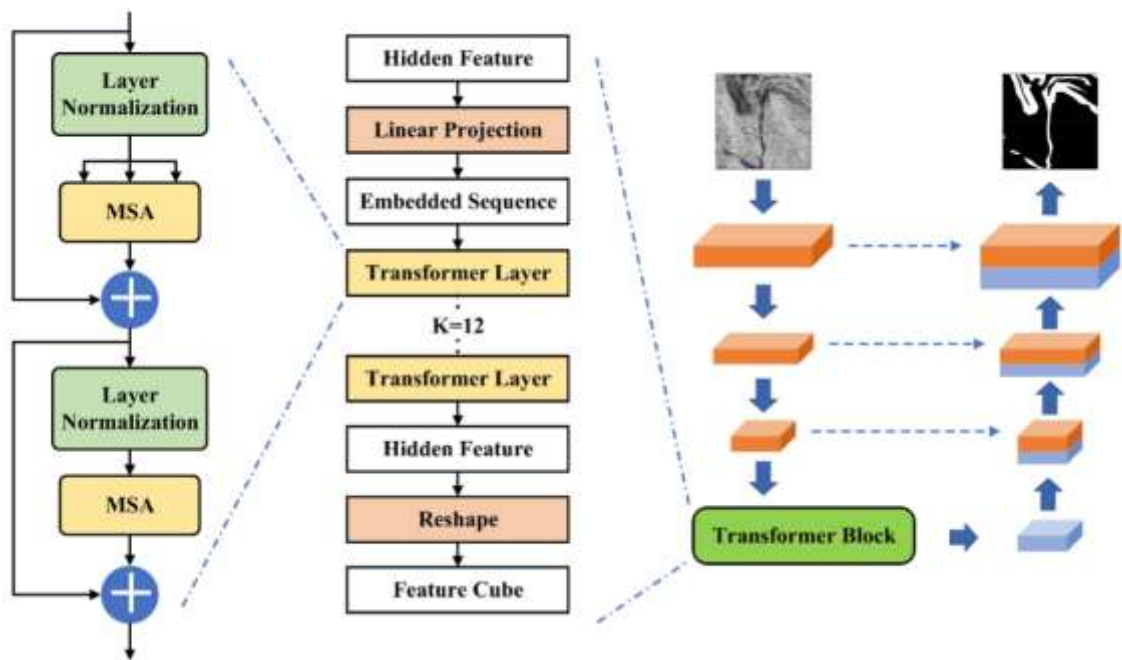
Outcomes: The goal is to develop an AI-powered system that:

- Automatically detects and segments oil spills in satellite images
- Learns visual patterns specific to oil-contaminated regions using deep learning
- Produces clear segmentation masks highlighting affected areas
- Assists environmental monitoring agencies in rapid oil spill detection and intervention
- Is deployable via a user-friendly interface for real-time use

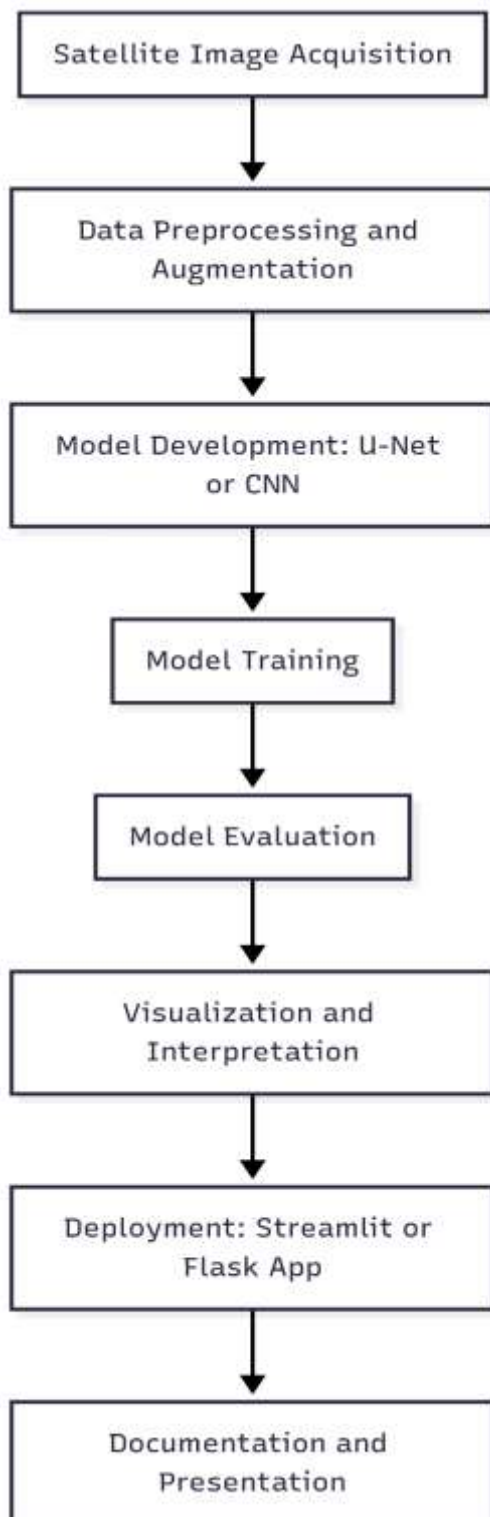
Modules to be implemented

- Data Collection
- Data Exploration (EDA) and Preprocessing
- Model Development (Segmentation and Classification)
- Training and Evaluation
- Visualization of Results
- Deployment via Streamlit or Flask App
- Presentation and Documentation

System Architecture for AI-Driven Oil Spill Detection and Monitoring



Workflow of the AI-Based Oil Spill Identification Process



Week-wise module implementation and high-level requirements with output screenshots

Milestone 1: Week 1-2

Module 1: Data Collection

- Acquire satellite image datasets such as Sentinel-1 SAR, MODIS, or NOAA.
- Dataset Used: [Oil Spill Detection Dataset on Kaggle](#)
- Obtain labeled oil spill annotations from open environmental datasets or manually create custom segmentation masks for training.
- Organize the dataset into a structured format with separate directories for training, validation, and testing.

Module 2: Data Exploration and Data Preprocessing

- Visualize sample satellite images and study statistical properties of spill and non-spill regions.
- Resize images to standard dimensions (e.g., 256x256).
- Normalize pixel values and apply SAR-specific filters such as speckle noise reduction.
- Augment dataset through techniques like flipping, rotating, scaling, and brightness/contrast variation.

Milestone 3: Week 3-4

Module 3: Model Development (Segmentation and Classification)

- Design and implement deep learning models like U-Net or CNN-based architectures.
- Customize input layers to handle single-channel SAR or multi-channel satellite data.
- Build the segmentation pipeline to predict oil spill regions from raw image inputs.

Module 4: Training and Evaluation

- Train the model using training datasets with real-time augmentation and validation.
- Implement loss functions like Dice Loss and Binary Cross-Entropy.
- Evaluate the model using metrics such as Accuracy, IoU (Intersection over Union), Dice Coefficient, Precision, and Recall.
- Fine-tune hyperparameters based on validation performance

Milestone 3 : Week 5-6

Module 5: Visualization of Results

- Generate side-by-side comparisons of original satellite images, ground truth masks, and predicted segmentation outputs.
- Overlay segmentation masks onto input images using color maps.
- Create visual summaries for use in reports and dashboards.

Milestone 4 :Week 7-8

Module 6: Deployment via Streamlit App

- Build a web-based interface that allows users to upload satellite images and view results in real-time.
- Implement backend model inference pipeline and frontend for user interaction.
- Optional: Integrate alerts, image storage, or API access for real-time monitoring systems.

Module 7: Presentation and Documentation

- Prepare comprehensive documentation including README files, architecture explanations, and usage instructions.
- Include model training guides, system diagrams, evaluation results, and visual output examples.
- Finalize project report and presentation slides for stakeholders or academic submission.

Evaluation Criteria:

Milestone 1 Evaluation (Week 1-2):

- Approval of dataset quality and organization.
- Approval of preprocessing and augmentation pipeline.
- Approval of raw vs. augmented image comparisons.

Milestone 2 Evaluation (Week 3-4):

- Approval of model architecture and implementation.
- Approval of loss function and optimizer settings.
- Approval of successful training loop and validation process.

Milestone 3 Evaluation (Week 5-6):

- Approval of evaluation metric results and comparison charts.
- Approval of overlay visualization and segmentation clarity.

Milestone 4 Evaluation (Week 7-8):

- Approval of working web interface for oil spill detection.
- Approval of user interface and result clarity.
- Approval of final documentation and presentation completeness.
- Final code submission with reproducible results and documentation.